Hyperbola

P(x,y) — M — S(α,β) → fx+my+n=0

For hyperbola, $e > 1$

→ It is locus of a pt. which moves in a plane such that ratio of its distance from a fixed point (s) to its dist. from a fixed line is always const. & greater than 1.

Condition on General 2nd degree Eqn :→

1) $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} \neq 0$

2) $h^2 - ab > 0$

Standard Eqn of Hyperbola :→

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

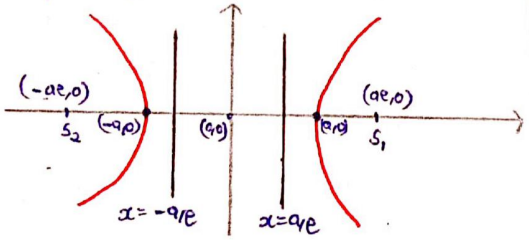
no condⁿ on y i.e. $y \in (-\infty, \infty)$

For y to exist

$(x^2 - a^2) \geq 0 \Rightarrow x \in (-\infty, -a] \cup [a, \infty)$

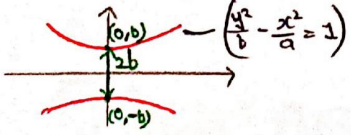
→ Means, no curve in $(-a, a)$ region

→ Graph is symmetric about x & y-axis.



* Focus : $(\pm ae, 0)$
* Vertices : $(\pm a, 0)$
* Directrix : $x = \pm a/e$
* Centre : $(0,0)$
* Conjugate Axis : 2b (Axis at which hyperbola does not intersect)
* Transverse Axis : 2a (Axis at which hyperbola intersect).

↳ Conjugate axis for $\left(\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1\right)$ is the transverse axis of its "conjugate hyperbola" i.e. $\left(\frac{-x^2}{a^2} + \frac{y^2}{b^2} = +1\right)$



$\left(\frac{y^2}{b} - \frac{x^2}{a} = 1\right)$

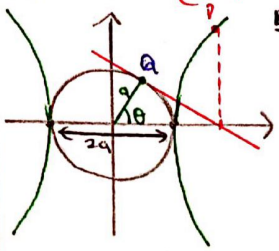
* Length of LR : $\frac{2b^2}{a}$ (b → conjugate axis)
* Eccentricity : $1 + \frac{b^2}{a^2}$
* Relation b/w eccentricities of Hyperbola and its conjugate :

$$\frac{1}{e^2} + \frac{1}{(e')^2} = 1$$

e → hyperbola ecc.
e' → C. hyperbola ecc.

#Auxiliary Circle and Eccentric Angle :

Circle drawn with centre C and Transverse axis as diameter is Auxiliary circle of hyperbola.



Eqⁿ of Auxiliary Circle : $x^2 + y^2 = a^2$

Parametric point : $Q \equiv (a\sec\theta, b\tan\theta)$

* P and Q are called "Corresponding Points" on the hyperbola.

#Alternate definition of Hyperbola :

It is locus of pt. which moves such that diff. of its distance from two fixed points (S_1 & S_2) always remains const. (length of TA) and this const. must be less than the distance b/w two fixed point ($2a < 2ae$)



Let, $|PS_1 - PS_2| = \lambda$

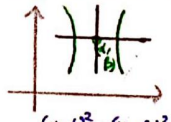
$\lambda < S_1S_2 \Rightarrow$ Locus is hyperbola
$\lambda = S_1S_2 \Rightarrow$ Locus is pair of Rays
$\lambda > S_1S_2 \Rightarrow$ No such point

#Position of point :

Let point be $P(x_1, y_1)$. Then if

$S_1 > 0 \Rightarrow$ Inside Hyperbola
$S_1 = 0 \Rightarrow$ On Hyperbola
$S_1 < 0 \Rightarrow$ Outside Hyperbola

#Hyperbola with shifted Centre :

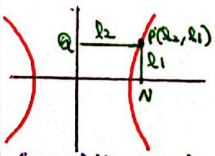


$H: \frac{(x-\alpha)^2}{a^2} - \frac{(y-\beta)^2}{b^2} = 1$



$H: \frac{(x-\alpha)^2}{a^2} - \frac{(y-\beta)^2}{b^2} = -1$

Important Point ⇒



$\frac{PQ^2}{a^2} - \frac{PN^2}{b^2} = 1$

$\frac{(\text{Var distance of P from CA})}{a^2} - \frac{(\text{Var distance of P from TA})}{b^2} = 1$

General Version ⇒

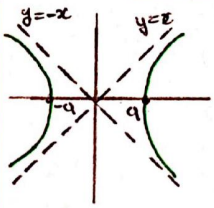
$lx + my + n = 0 \longrightarrow$ Eqⁿ of TA, $2a \rightarrow$ Length of TA
$mx + ly + p = 0 \longrightarrow$ Eqⁿ of CA, $2b \rightarrow$ Length of CA

$$\therefore H = \frac{\left(\frac{mx+ly+p}{\sqrt{l^2+m^2}}\right)^2}{a^2} - \frac{\left(\frac{lx+my+n}{\sqrt{l^2+m^2}}\right)^2}{b^2} = 1$$

#Rectangular Hyperbola :

Hyperbola, whose TA = CA, i.e. $a = b$, $e = \sqrt{2}$

$\Rightarrow H: x^2 - y^2 = a^2$

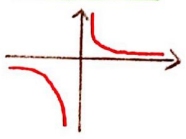


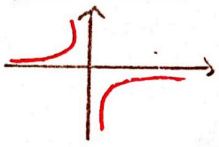
$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \rightarrow P(a\sec\theta, b\tan\theta)$

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1 \rightarrow P(a\tan\theta, b\sec\theta)$

* Equation of Hyperbola w.r.t its asymtotes as coordinate axes is.

$H: xy = c^2$ OR $H: xy = -c^2$





* Eccentricity = $\sqrt{2}$
* Assymptotes ⇒ $y = 0$ and $x = 0$
* TA ⇒ $y = x, 2\sqrt{2}c$; CA ⇒ $y = -x, 2\sqrt{2}c$
* Vertex ⇒ $A(c, c)$ & $A'(-c, -c)$
* Foci ⇒ $S(\sqrt{2}c, \sqrt{2}c)$ & $S'(-\sqrt{2}c, -\sqrt{2}c)$
* Length of LR = $2\sqrt{2}c$
* Auxiliary Circle : $x^2 + y^2 = 2c^2$
* Eqⁿ of Directrix : $x + y = \sqrt{2}c$, $x + y = -\sqrt{2}c$